MMF 1921 Exam May 12, 2025

Instructions: Closed book and closed notes except for one side of a 8.5 by 11 inch sheet of handwritten notes. Only a simple scientific non-financial calculator with no programming capability is allowed. Write **neatly** as this will aid in providing maximum partial credit. **Show All Work.**

IMPORTANT: Interpretation of the exam questions is part of the exam and so no questions will be taken DURING the exam that ask for clarification of the question. You MUST turn in this question sheet with your answer booklet or else your exam will NOT be marked.

A useful fact:

If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and is invertible, then $A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

Problem 1 (8 points)

You currently have a porfolio of eight stocks with weights given in the table below. Suppose you also used Markowitz optimization and computed an optimal porttfolio over the eight stocks with weights given in the table below.

stock	1	2	3	4	5	6	7	8
your portfolio	0.12	0.15	0.13	0.10	0.20	0.10	0.12	0.08
Markowitz	0.02	0.05	0.25	0.06	0.18	0.10	0.22	0.12

You would like to rebalance your portfolio in order to be closer to the mean-variance portfolio. To avoid excessively high transaction costs, you decide to rebalance on only three of the stocks from your portfolio. The objective is to minimize the quantity

$$|x_1 - 0.02| + |x_2 - 0.05| + |x_3 - 0.25| + \cdots + |x_8 - 0.12|$$

which measures how closely the rebalanced portfolio matches the mean-variance portfolio.

Formulare this rebalancing problem as an optimization problem and formulate the most tractable version (this means the formulation would be as amenable as possible to being solved directly by a relevant algorithm). Write out the model in complete detail (e.g. show variables with appropriate subscripts) and define all variables. Do not use summation notation.

Problem 2 (22 points)

Consider the (primal) LP

$$\begin{array}{ll} \text{maximize} & 3x_1 + 5x_2 + 2x_3 \\ \text{subject to} & 2x_1 + x_2 + x_3 \leq 10 \\ & x_1 + 3x_2 + 2x_3 \leq 15 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0 \end{array}$$

(a) (3 points) Write the dual of this LP.

(b) (10 points) Solve the (primal) LP using the simplex method (you must use the version developed in class).

(c) (3 points) Use your work from (b) to show that at termination of the simplex method the dual problem is solved to optimality as well.

(d) (3 points) Show that the search direction d you obtain after the first iteration of the simplex method in this problem satisfies $Ad = 0$. Explain why having d satisfying this condition is important.

(e) (3 points) Show that the dual solution after the first iteration is infeasible for the dual but that complementary slackness holds.

Problem 3 (26 points)

Part 1

Consider the function $f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$.

(a) (3 points) Prove $x = \binom{1}{1}$ is the unique global minimizer of $f(x)$ over $\mathbb{R}^2$.

(b) (8 points) With an initial point $x^{(0)} = \binom{0}{0}$ apply two iterations of pure Newton's method for minimizing $f(x)$ over $\mathbb{R}^2$.

Part 2 (3 points each)

Let $f(x) : \mathbb{R}^1 \to \mathbb{R}^1$ be given by $f(x) = (x - x_0)^4$ where x_0 is a constant. Suppose that we apply Newton's method to minimizing $f(x)$.

(a) Write the Newton recurrence (i.e. the formula for $x^{(k+1)}$) for Newton's method applied to this problem.

(b) Let $y^{(k)} = |x^{(k)} - x_0|$, where $x^{(k)}$ is the kth Newton iterate. Show that the sequence $\{y^{(k)}\}$ satisfies $y^{(k+1)} = \frac{2}{3}y^{(k)}$.

(c) Show that the sequence $\{x^{(k)}\}$ converges to x_0 for any initial guess $x^{(0)}$.

(d) Show that the sequence $\{x^{(k)}\}$ converges linearly.

(e) Why does the sequence $\{x^{(k)}\}$ not converge quadratically?

Problem 4 (8 points)

Consider the following variant of mean-variance portfolio optimization for constructing a porfolio of n risky assets and one risk free asset with return r_f where one maximizes risk adjusted return subject to only a budget constraint with short-selling allowed i.e.

$$\begin{array}{ll} \min\limits_{x,y} & \mu^T x + r_f(x_{n+1}) - \frac{1}{2}\gamma x^T Q x \\ \text{s.t.} & 1^T x + x_{n+1} = 1 \end{array}$$

where μ is the expected return vector, Q is the covariance matrix and $\gamma > 0$ is a risk-aversion coefficient. The vector x contains the weights of the n risky assets and x_{n+1} is the weight for the risk free asset.

(a) (6 points) Find a closed-form solution for the problem above. State the conditions under which the closed-form solution is well defined and unique and justify your answer.

(b) (2 points) Does the closed-form solution in (a) have any interpretation? Explain BRIEFLY why or why not.

Problem 5 (8 points) True or False

You are analyzing an investment opportunity with two stocks A and B with the expected return of Stock A $\mu_A = 0.025$, the expected return of Stock B $\mu_B = 0.015$, the variance of Stock A $\sigma_A^2 = 0.0049$, the variance of Stock B $\sigma_B^2 = 0.0025$, and $\sigma_{AB} = -0.001$.

Suppose that the expected returns were estimated using the last 20 monthly observations. Now consider the following robust optimization problem with an ellipsoidal uncertainty set

$$\begin{array}{ll} \min\limits_{x,y} & x^T Q x \\ \text{s.t.} & \mu^T x - y \geq 1\% \\ & y^2 \geq x^T \Theta x \\ & 1^T x = 1 \\ & y \geq 0 \end{array}$$

Use this information to check whether the solution $x = [0.425\ 0.575]^T$ satisfies the KKT conditions of the problem above. (Note: Assume the target return constraint is active.)

Problem 6 (10 points)

(a) (3 points) Consider three risky assets with the following covariance matrix

$$Q = \begin{bmatrix} 0.0144 & 0.0144 & 0.0012 \\ & 0.04 & 0.005 \\ & & 0.0025 \end{bmatrix}$$

Now consider the following portfolio of the three assets given by the vector

$$x = \begin{bmatrix} 0.35 \\ 0.10 \\ 0.55 \end{bmatrix}$$

Determine whether the vector x is a risk parity portfolio.

(b) (3 points) Pick the least complex (simplest) risk parity model that is based on a least squares objective and formulate that version of the model for the covariance matrix in part (a). Make sure that in your formulation to expand all summations and write the model in detail using actual numbers for coefficients.

(c) (4 points) Prove that your model in (b) is a non-convex optimization model.

Problem 7 (8 points, 2 points each) True or False (briefly justify your choice)

(a) Under the One Fund Theorem all investors invest in the same risky asset in the same amount.

(b) The role of γ in function $F_\alpha(x, \gamma)$ is to be a place holder (variable) that will ultimately be the value of VaR_α.

(c) In the Black-Litterman framework subject views are assumed to by statistically dependent.

(d) In the Black-Litterman framework the expected return vector that is generated is obtained only through reverse optimization.

Problem 8 (10 points)

Consider the problem

$$\begin{array}{ll} \text{minimize} & f(x) + g(x) \\ \text{s.t.} & Ax = b \\ & x \geq 0 \end{array}$$

where $f(x)$ and $g(x)$ are differentiable functions from $\mathbb{R}^n$ to $\mathbb{R}^1$.

(a) (3 points) Formulate the barrier problem for the problem above and derive the KKT conditions

(b) (4 points) Using your results from (a) write out the system of linear equations that when solved would generate the search direction for a primal-dual interior point method for the problem above.

(c) (3 points) What are the requirements on the functions $f(x)$ and $g(x)$ in order for the Newton system in (b) to be well-defined and a decent direction?